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COS 484: Natural Language Processing

Paper: The Reversal Curse: LLMs Trained on 'A is B' Fail to Learn 'B is A' - Berglund et al., 2023

Overview: The Reversal Curse (Berglund et al., 2023) demonstrates a fundamental asymmetry in how autoregressive language models encode factual knowledge. A model trained on the statement “Tom Cruise's mother is Mary Lee Pfeiffer” does not automatically learn the logically implied reverse “Mary Lee Pfeiffer's son is Tom Cruise”. The model's forward recall is near-perfect, but its reverse recall is near-random. In our study, we aim to test the Reversal Curse on modern models (for example, with instructing tuning or chain-of-thought prompting). We also test whether the Curse compounds over multiple hops or extends to other relationships.

Link to paper: <https://arxiv.org/abs/2309.12288>

Codebase: https://github.com/lukasberglund/reversal_curse

Baselines

- Fine-tune GPT-2 Small/Medium/Large on the paper's synthetic celebrity-description dataset (~1,000 name-description pairs). Evaluate forward accuracy ($A \rightarrow B$) and reverse accuracy ($B \rightarrow A$)
- Reproduce the GPT-4 evaluation (via API) on the real-world celebrity name-description pairs as an upper-bound comparison
- Fine-tune LLaMA on QuestionToAnswer vs. AnswerToQuestion instruction formats $\rightarrow$ does direction of instruction matter?

New Baselines

- Fine-tune matched-size modern language models (SmolLM2-135M, 360M, and 1.7B), possibly with LoRA on the same synthetic celebrity-description dataset and compare forward vs. reverse accuracy against GPT-2 Small/Medium/Large to test whether recent architectural improvements and better pretraining data alleviate the reversal curse.
- Reproduce the real-world celebrity name-description pairs using GPT 5.3 and 5.4 to check for improvements
- Multi-Hop Reversal Baseline: Test whether reversal curse compounds over hops. If A is B and B is C is learned, can the model answer C is A?
- Bidirectional training baseline: include both 'A is B' and 'B is A' in the training set for a k% subset ($k=0, 10, 25, 50, 75, 100$) and measure how much reverse accuracy is restored.

Ablation Studies

- Training Repetition Frequency Ablation: Repeat each training example R times where R in $\{1, 5, 20, 50, 100\}$ and test if frequency of exposure to the forward direction makes the curse worse?
- Paraphrase probing: at evaluation time, rephrase the reverse query in different surface forms. E.g. original/fill in the blanks/indirect question/possessive reversal/yes or no verification to test if the model has the knowledge but fails at a specific surface form.
- Instruction Tuning Ablation: at inference time, test using chain of thought/zero-shot direct/few shots prompting and check if that recovers reverse accuracy.
- Relation type generalization: extend beyond 'is' relationships. Create synthetic datasets for 'X was born in Y', 'X wrote Y', and 'X is the capital of Y' — does the reversal curse hold equally across relation types like birth location, awards, capital etc.

Compute and Resource Estimate

- Synthetic celebrity-description dataset generation ($\sim 1,000$ pairs) and multi-hop chain dataset (~ 800 training pairs): ~ 10 minutes total on CPU.
- GPT-2 Small fine-tuning (1,000 examples, 100 epochs, 3 seeds): $\sim 10\text{--}15$ min/run on a T4 GPU. Total: $\sim 30\text{--}45$ min.
- GPT-2 Medium fine-tuning (same setting, 3 seeds): $\sim 25\text{--}35$ min/run on T4. Total: $\sim 75\text{--}105$ min.
- GPT-2 Large fine-tuning (same setting, 3 seeds): $\sim 50\text{--}70$ min/run on T4. Total: $\sim 2.5\text{--}3.5$ GPU-hours.
- SmolLM2-135M fine-tuning (100 epochs, 3 seeds): $\sim 12\text{--}18$ min/run.
SmolLM2-360M: $\sim 28\text{--}38$ min/run. SmolLM2-1.7B: $\sim 60\text{--}80$ min/run. Total SmolLM2 suite: $\sim 5\text{--}7$ GPU-hours.
- LLaMA-2 7B with QLoRA fine-tuning for QuestionToAnswer vs. AnswerToQuestion (1,000 examples, 20 epochs, 2 seeds each): $\sim 30\text{--}45$ min/run on T4. Total: $\sim 2\text{--}3$ GPU-hours.
- GPT-5.3 Instant and GPT-5.4 Thinking API evaluation on real-world celebrity pairs (~ 430 pairs, zero-shot): $\sim \$0.10$ for GPT-5.3 Instant; $\sim \$2\text{--}3$ for GPT-5.4 Thinking. No GPU required.
- Bidirectional training baseline ($k = 0, 10, 25, 50, 75, 100$ on GPT-2 Medium, 2 seeds): $\sim 20\text{--}25$ min/run. Total: $\sim 4\text{--}5$ GPU-hours.
- Multi-hop reversal baseline (GPT-2 Medium, 2 seeds): $\sim 20\text{--}25$ min/run. Total: $\sim 45\text{--}60$ min.
- Training Repetition Frequency ablation ($R = 1, 5, 20, 50, 100$ on GPT-2 Medium, 2 seeds): $\sim 3\text{--}4$ GPU-hours.

- Relation type generalization ablation (4 relation types, GPT-2 Medium, 2 seeds): ~2.5–3.5 GPU-hours.
- Paraphrase probing and Instruction Tuning ablations (inference only, all fine-tuned models): ~1–2 GPU-hours combined.
- Total estimated compute: ~25–35 GPU-hours. Recommended: Google Colab Pro (\$10/month, ~80 T4 GPU-hours). API costs: ~\$3–5 total for GPT-5.3 and GPT-5.4 evaluations.

Proposed Timeline

- Week 1: Generate synthetic celebrity-description and multi-hop chain datasets (all members, CPU). Set up the environment and download model weights. Reproduce GPT-2 Small, Medium, and Large baselines with 3 seeds each (Member A & B). Begin SmoLM2-135M and 360M fine-tuning (Member C).
- Week 2: Complete SmoLM2-360M and 1.7B fine-tuning; compile GPT-2 vs. SmoLM2 comparison table (Member A & B). Fine-tune LLaMA-2 7B with QLoRA on QuestionToAnswer and AnswerToQuestion instruction datasets (Member C).
- Week 3: Run GPT-5.3 Instant and GPT-5.4 Thinking API evaluation on real-world celebrity pairs (Member A). Run bidirectional training baseline across all k values on GPT-2 Medium (Member B). Run multi-hop reversal baseline: fine-tune and evaluate 1-hop vs. 2-hop accuracy (Member C).
- Week 4: Training Repetition Frequency ablation ($R = 1, 5, 20, 50, 100$) on GPT-2 Medium (Member A). Relation type generalization: create and fine-tune on born_in, wrote, capital datasets (Member B). Paraphrase probing: evaluate all models on 5 reverse query surface forms (Member C).
- Week 5: Instruction Tuning Ablation: test CoT, zero-shot, and few-shot prompting on fine-tuned models (Member A & B). Compile and interpret all results; begin the paper draft covering introduction, related work, and experimental setup (Member C).
- Week 6: Complete paper draft including results, discussion, and conclusion sections (all members). Finalize all figures and tables. Prepare and rehearse the presentation (all members).